

tached to the heat-bed using aluminium tape. The thermistor should poke through the central hole in the bed, about 1mm. Tape the thermistor securely and run the cable to the right (Fig. 12). The cable is connected to Sanguinololu board for monitoring the temperature of the heat-bed.

Extruder. The extruder's nozzle end heats up when enabled by the Sanguinololu board. The extruder assembly has an in-built thermistor to monitor and control the temperature of the hot end. A stepper motor rotates to push the raw material in the nozzle from the cold end; the speed can be controlled. The material melts when pushed through the nozzle's hot end and gets placed on the heat-bed as per 3D model design. Fig. 13 shows the nozzle (extruder) assembly.

Software

All the software required for a DIY type 3D printer is available at:

<http://www.nextdayreap.co.uk/reap-downloads/>

The Sanguinololu board comes with boot-loader preinstalled. You just have to upload the 3D printer firmware. For that, first install Arduino development environment on your computer and add the Sanguinololu board in it. Download Sanguino software from: <https://code.google.com/p/sanguino/downloads/list>

Copy the downloaded software in the Arduino installation directory under Hardware folder. The Sanguino boards will start reflecting in the boards menu of Arduino IDE. Now, select the board in the Arduino IDE corresponding to the microcontroller on the Sanguinololu board. Download the 3D printer firmware (Marlin or Sprinter), open it in the IDE and click on Verify/Compile button. Click on Upload button to upload the firmware in the Sanguinololu board. Note that you might have to make some changes in the Configuration.h file as per your printer.

Once the firmware is uploaded, you just need software that can slice the 3D model design (.STL file) to make the .gcode file. This .gcode file is

then given to the 3D printer for printing the object, layer by layer. The software that you can use here is Pronterface. Follow the steps below to install Pronterface and dependencies.

1. Install Python Environment & Dependencies. Download and install the 'python-2.7.2.msi' from <http://python.org/ftp/python/2.7.2/python-2.7.2.msi>

2. Download and install PYSerial for serial communication from <http://pypi.python.org/packages/any/p/pyserial/pyserial-2.5.win32.exe>

3. Download and install Python 8 for Python from <http://downloads.sourceforge.net/wxpython/wxPython2.8-win32-unicode-2.8.12.0-py27.exe>

4. Download and install PYReadline also for serial communication from <http://launchpad.net/pyreadline/1.7.1.7/+download/pyreadline-1.7.win32.exe>

5. Now, install Pyglet. Download it from the link <http://pyglet.googlecode.com/files/pyglet-1.1.4.zip>

6. Extract the files. Run command prompt and navigate to the extracted directory. Type 'setup.py install' and press enter as shown in Fig. 14.

7. Download and install Pronterface from www.nextdayreap.co.uk/downloads/klimet-Printrun-d9a3363.zip

Mechanical assembly

Mechanical assembly is the most critical thing in DIY type 3D printers as all the parts need to work precisely to print an object properly. Mechanical assembly instructions and architecture will differ from model to

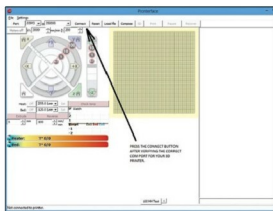


Fig. 15: Setting COM port and baud rate

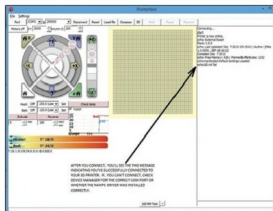


Fig. 16: Successful connection to 3D printer

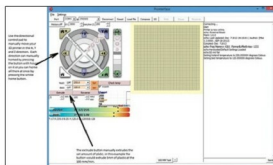


Fig. 17: Manual control of 3D printer

